Neural correction trees: augmentation and generalization

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Completed 23 October 2026: **166 primary configurations + 44 selected repeats = 210 evaluated configurations.** All final test evaluations are saved. Continued epoch budgets are configurations, not independent training runs.

What we investigated

We extended the earlier error-correction tree with image augmentation, majority-vote ensembles, sample/epoch bias-variance comparisons and local loss-landscape plots. The aim was to understand when the tree helps, when it memorizes, and whether a second descent appears after training error reaches zero.



Frozen root score + positive child correction - negative child correction

Children are corrected recursively; only correction networks and signed strengths learn.

How the design developed

Earlier experiment	Reason for trying it
Trained shared CNN	Efficient useful features, but the strong root could hide the tree contribution.
Pixels / selective local CNN nodes	Give more learning responsibility to the tree; retain matched ordinary-classifier controls.
CNN stopped near 80%; random CNN	Leave mistakes for corrections; separately remove supervised feature learning.
15% training-label noise	Study harder fitting and distinguish wider-root gains from correction-tree gains.
This follow-up: fixed input views	Make augmentation reproducible and interpolation measurable; add voting and analysis.

Data and common settings

Dataset	Train / validation / test	Finite training views
CIFAR-10	45,000 / 5,000 / 10,000	135,000 with augmentation
Cats vs Dogs	18,608 / 2,326 / 2,326	55,824 with augmentation

Both datasets use 32 x 32 inputs and the saved CNN80 features. A frozen CNN/head supplies the root decisions; correction MLPs use 128 features. CIFAR uses ten one-versus-rest trees and Cats uses one binary tree. Adam uses learning rate 0.001; no dropout, weight decay or label smoothing. Main fits allow 400 epochs per child. Widths span 16-512; depth limits are 0/1/2/4/8/12, with at most 127 nodes per binary tree. Hardware: RTX 2050, 4 GB GPU memory.

1. Augmentation and memorization

For each original image we cache its unchanged view plus two reflection-padded random crops with horizontal flips. Image ID, view number and seed fix each transform. A separate control repeats the original three times. This matches image-view exposure, but child masks can change optimizer-update counts. These are not three independent images.

Dataset	Input views	Original train %	All-view train %	Val %	Test %
CIFAR	Original	94.46	94.46	76.50	76.13
CIFAR	Repeated	99.26	99.26	74.78	74.44
CIFAR	Augmented	90.67	88.33	77.42	76.69
Cats	Original	84.25	84.25	81.47	81.17
Cats	Repeated	86.50	86.50	80.91	80.87
Cats	Augmented	83.74	81.99	81.30	81.64

These are the predeclared width-64, depth-one reference models. Augmentation raises test accuracy by **0.56 points on CIFAR** and **0.47 on Cats** over the original-only reference. The gains are small descriptive comparisons, not a significance claim. Repeating originals improves fitting while lowering test accuracy.

The frozen CNN alone scores **75.70% CIFAR and 82.07% Cats test accuracy**. The Cats reference trees do not beat that baseline. Correction-tree memorization therefore must not be presented as a consistent improvement in generalization.

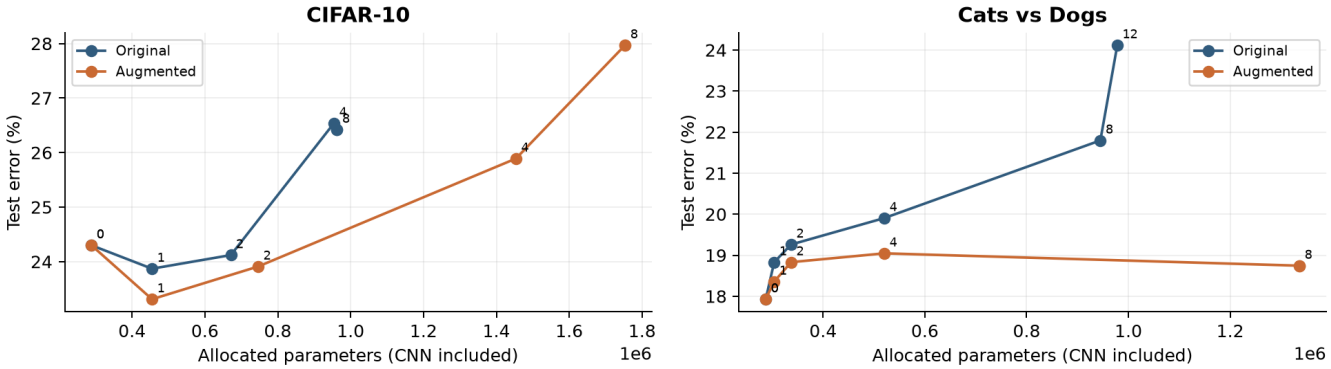


Figure 1. Fixed-root depth study, width 64. Point labels are requested depth limits, not actual depths. Lines join measured models; repeated sizes can overlap.

Dataset	Views	Limit / actual depth	All-view train %	Test %
CIFAR	none	8 / 5	100.00	73.58
CIFAR	augmented	8 / 6	100.00	72.03
Cats	none	12 / 10	100.00	75.88
Cats	augmented	12 / 8	88.17	81.26

The tree can fit beyond the imperfect frozen root. However, deeper fitting mostly worsens test performance. Cats with augmentation still has 6,605 training-view errors. Depth limits 8 and 12 produce identical CIFAR models; requesting a deeper limit does not necessarily add capacity.

2. Did we observe double descent?

The relevant outcome is held-out error as capacity grows. Training errors locate interpolation: exact zero classification errors on the actual fitting targets and views. A second descent is a later fall in held-out error around/beyond that transition [2]. Its peak need not land exactly on the first sampled zero-error point.

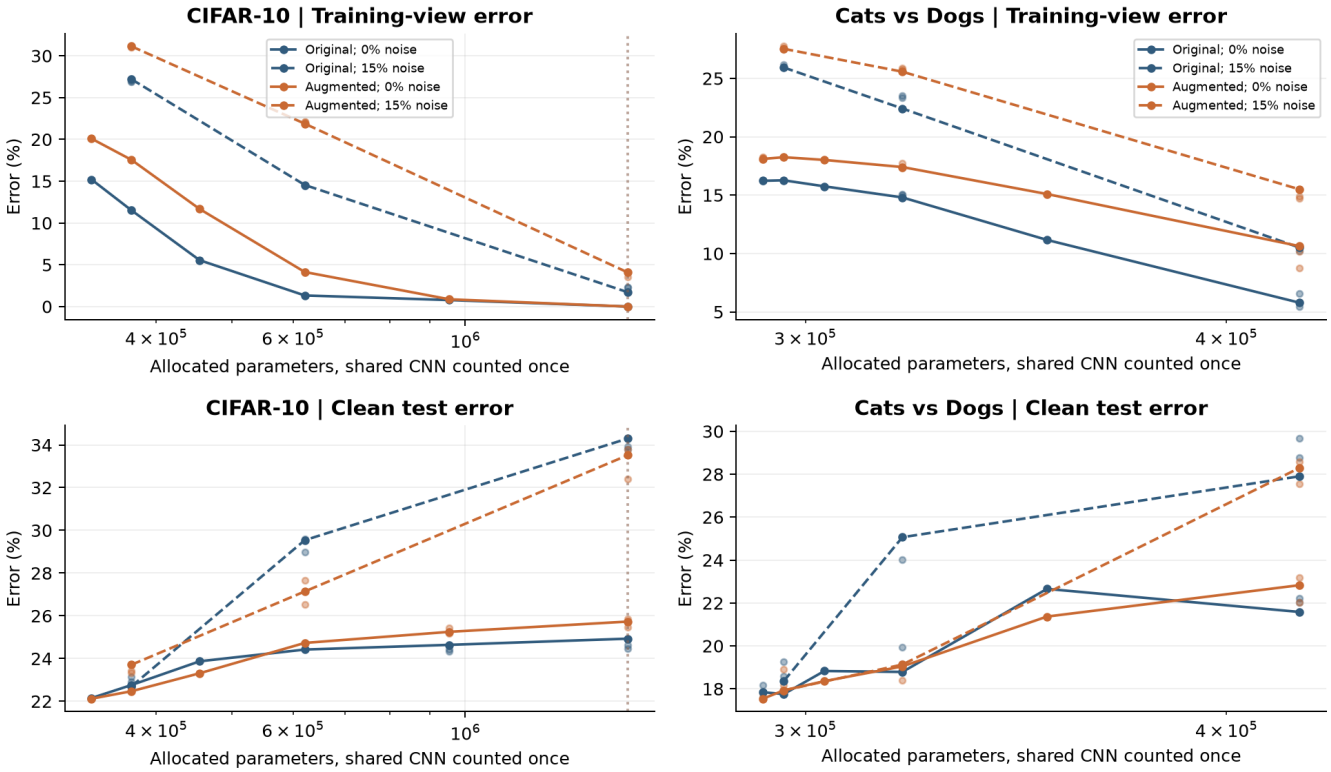


Figure 2. Current width study: primary seed 409 curves, with extra optimizer-seed results as faint dots at selected widths. Dotted lines mark first tested zero-error models. Test labels remain clean; 15% noise changes training targets only.

Current result: robust double descent is not established. Clean CIFAR reaches zero training errors at width 512 in all three seeds, but we did not test larger widths. Noisy CIFAR still has errors at width 512: 771-1,051 without augmentation and 4,742-5,535 with it. Cats does not interpolate in this width sweep. The completed range therefore does not adequately cover several capacities beyond interpolation.

There are small fluctuations: original-view CIFAR depth 4 to depth 8 gains 0.12 test-accuracy points, and original-view Cats width 256 to 512 also improves. The former is a single-seed comparison and the latter does not interpolate. Neither establishes a repeatable post-interpolation tree effect.

Earlier label-noise evidence - a separate experiment

Earlier CIFAR width	Zero-error runs, 1,200 epochs	Mean test error %
256	3/3	30.31
512	3/3	28.09
1024	3/3	26.36

Earlier V5 used learned MLP roots, unlike the frozen CNN roots here. At 400 epochs its noisy CIFAR test error fell, rose and fell again: 32.60% / 23.43% / 30.21% / 26.29% at widths 2/8/128/1,024. The longer follow-up above confirms later improvement among interpolating models, but those wide models had inactive correction children. This is primarily a root-width result, not evidence that growing the tree caused double descent.

3. Multiple trees and majority voting

For each dataset, augmentation arm and sample/epoch setting, we trained three stratified bootstrap members. All views of a sampled original stay together. Each complete classifier contributes one final class vote, following bagging [3]. A CIFAR member contains ten binary trees; those are not ten ensemble votes. Ties choose the smallest class index.

Dataset	Views	Epochs	First member %	3-member vote %	Change, pp
CIFAR	Original	20	77.50	77.63	+0.13
CIFAR	Original	100	76.84	77.53	+0.69
CIFAR	Original	400	75.14	76.62	+1.48
CIFAR	Augmented	20	77.90	77.93	+0.03
CIFAR	Augmented	100	75.71	76.90	+1.19
CIFAR	Augmented	400	74.49	75.87	+1.38
Cats	Original	20	81.77	81.99	+0.21
Cats	Original	100	82.07	82.24	+0.17
Cats	Original	400	76.83	77.09	+0.26
Cats	Augmented	20	81.81	81.94	+0.13
Cats	Augmented	100	82.12	82.24	+0.13
Cats	Augmented	400	78.55	79.02	+0.47

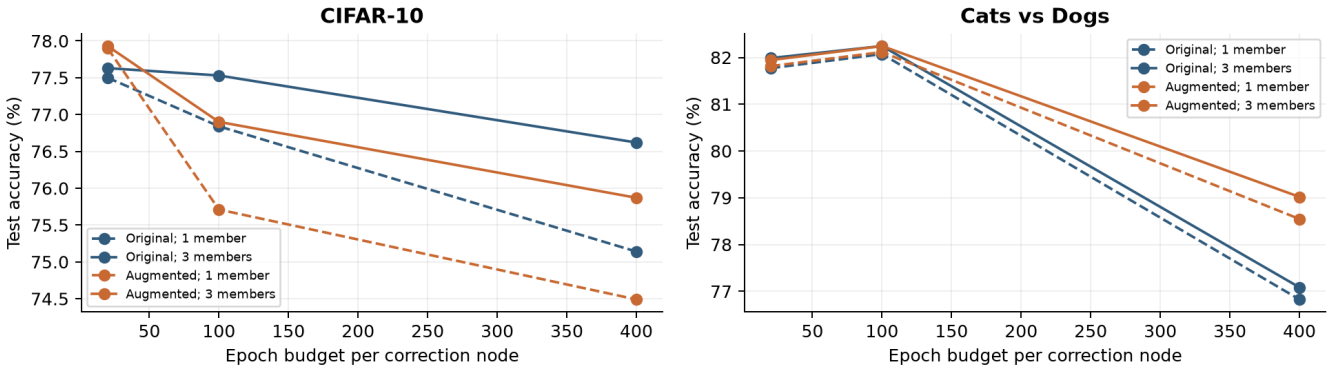


Figure 3. Test accuracy of one predeclared bootstrap member versus three-member voting. Settings share the same feature extractor; this is not a comparison to the separate full-data reference model.

Voting improves on the first member in all twelve comparisons, although some gains are tiny. At 400 epochs the CIFAR gains are 1.48 points without augmentation and 1.38 with it. Nevertheless, long-trained voting models remain below some earlier-budget models: averaging does not remove all overfitting.

Only three members were used, and they share the CNN. Voting reduces some member-specific mistakes; it cannot remove errors shared by every member. Bootstrap members differ in sampled originals and initialization. Width-study optimizer repeats instead keep the original fitting set fixed.

4. Bias and variance across samples and epochs

For each held-out validation image, let p_m be member m's class-probability vector and y its observed one-hot label. We report the exact empirical Brier identity:

mean ||p_m - y||^2 = ||mean(p_m) - y||^2 + mean ||p_m - mean(p_m)||^2.

The two terms are a **squared-bias proxy** against the observed labels and **bootstrap prediction variance**. They are conditional on the shared CNN already trained on the full clean pool. This does not estimate Bayes bias, irreducible noise or full-model sample complexity. Three members give limited replication.

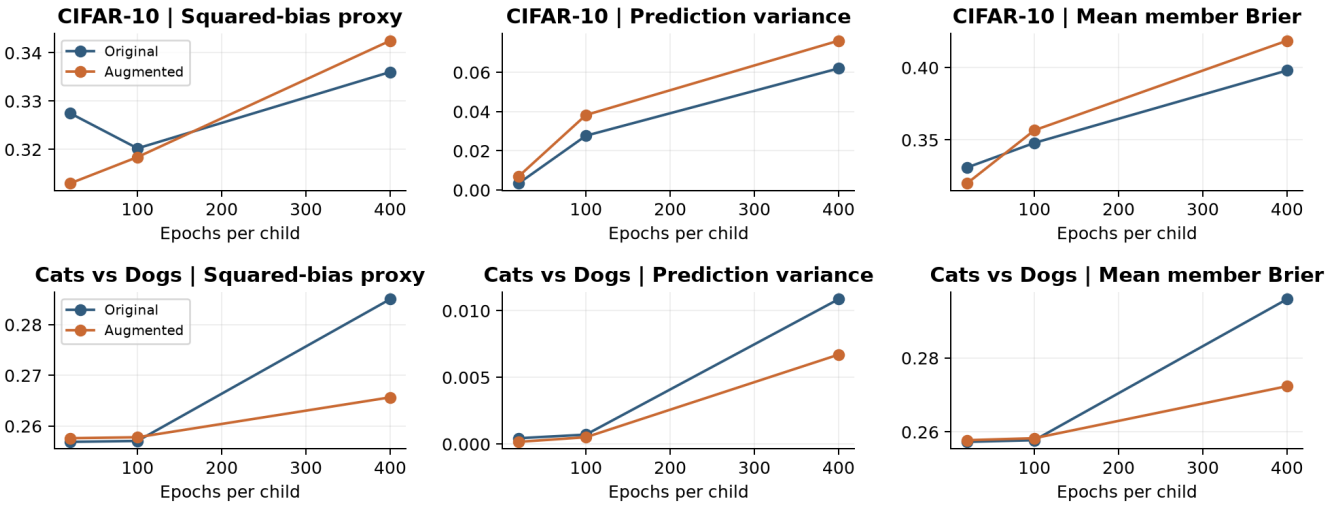


Figure 4. Full-size bootstrap draws, three members. Longer per-child budgets generally raise prediction variance; the squared-bias proxy need not move in the opposite direction.

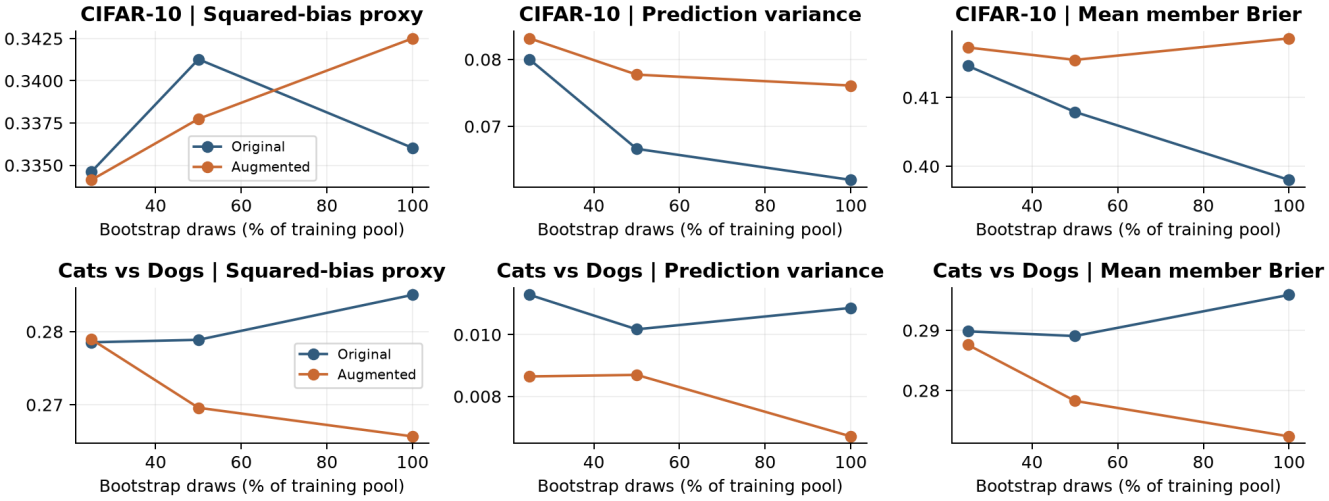


Figure 5. Sample-size comparison at 400 epochs per child. Percentages count original-image draws with replacement, not unique images or augmented views.

For original-view CIFAR, variance rises from **0.00343 at 20 epochs to 0.06199 at 400**. With augmentation it rises from 0.00699 to 0.07610. At 400 epochs, increasing draw count from 25% to 100% lowers original-view CIFAR variance from 0.0800 to 0.0620; at 20 epochs it instead rises slightly from 0.0025 to 0.0034. More data does not uniformly reduce every measured term.

For Cats at 400 epochs, augmentation lowers variance from 0.01086 to 0.00671 and the squared-bias proxy from 0.28502 to 0.26566. Augmentation can help both terms in this setting. Across CIFAR, the same improvement is not universal. All 36 decomposition groups satisfy the empirical identity within numerical tolerance.

5. What the loss landscapes show

For each predeclared reference model, we evaluate a 15 x 15 grid in two random parameter directions, normalized by the output-row norms of MLP weight matrices [4]. The CNN, frozen root, biases and correction strengths stay fixed. We recompute hard correction decisions on every grid point.

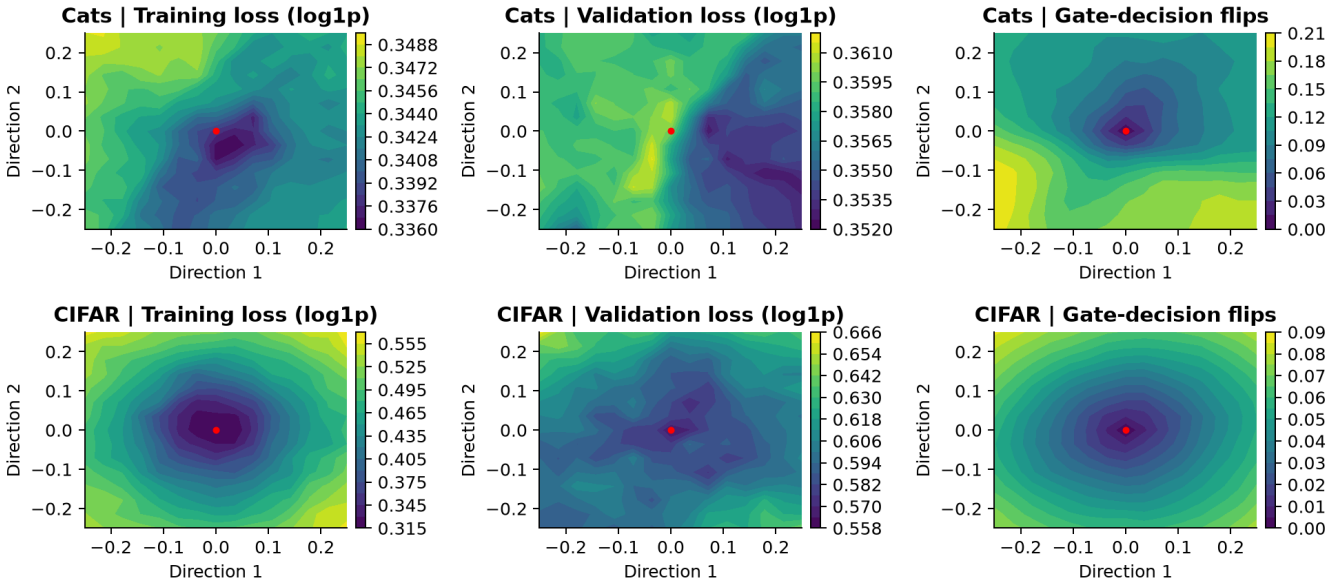


Figure 6. Augmented reference models. The red point is the saved model. The first two columns show log1p final classification loss on 512 training/validation examples; the third shows changes in node gate decisions.

Dataset	Center train loss	Center val loss	Lowest val loss on grid	Max gate flips
Cats	0.4023	0.4332	0.4226	20.70%
CIFAR	0.3706	0.7555	0.7555	8.62%

The Cats slice has lower validation loss away from the saved point (0.4226 versus 0.4332 at its center). Its gate decisions can change by 20.70% on this grid. For augmented CIFAR, the center attains the lowest measured training and validation loss on the grid; up to 8.62% of gate decisions change under perturbations. The percentage counts node decisions, not images whose final class changes.

These observations show local sensitivity and the effect of discrete corrections. They do not prove that one model has a globally flatter or better minimum. The plotted final-classifier loss differs from the separate masked objectives used to train nodes, so the saved model need not be stationary for this plot. Contour interpolation is a visualization of the grid; hard gates can create discontinuities.

Implementation checks

The pipeline was checked for target masks/signs, grouped augmentation sampling, exact-task optimizer/RNG continuation, changed-task rejection, voting, Brier accounting, deployment round trips and landscape restoration. Eleven checks and a separate 58-configuration small pilot passed. Pilot accuracy is not full-data evidence. All 210 planned configuration IDs, test evaluations and parameter sums were audited from the final saved results.

6. Additional checks: views and useful corrections

We tested an additional prediction rule without fitting new weights: each validation image contributes its original view plus seven deterministic crop/flip views, and all member/view class predictions vote. The same view bank is used for every model. Ties choose the smallest class index. These are exploratory validation results, not new test scores.

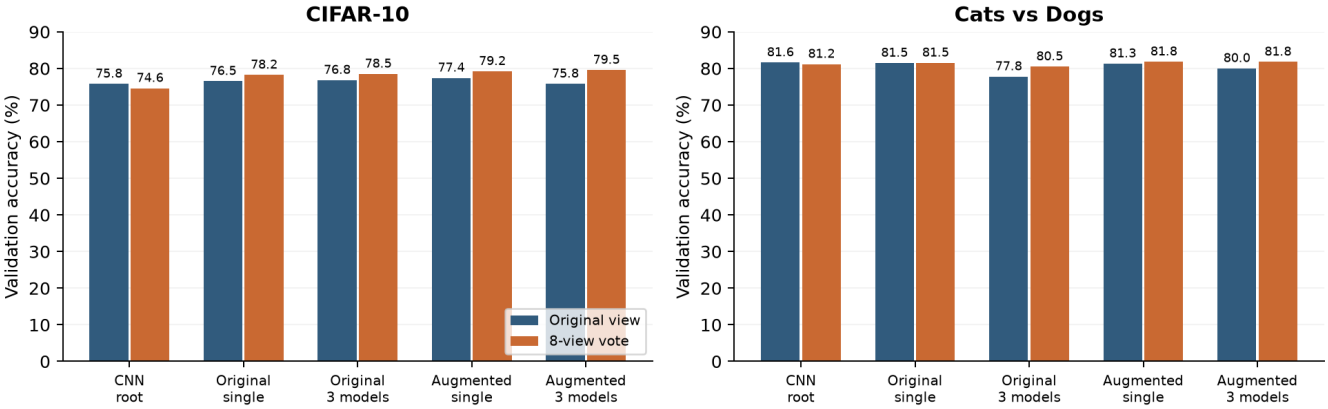


Figure 7. Eight-view voting using unchanged saved deployments. Each CIFAR ensemble member is a complete ten-tree classifier. The CNN root is a separate control.

For CIFAR, the augmented single reference improves from 77.42% to 79.20%, and the augmented three-member ensemble from 75.78% to 79.54%. The frozen root falls from 75.76% to 74.60%. View voting therefore helps these correction models even though it does not improve the root. For Cats, the augmented single improves from 81.30% to 81.81% and the ensemble from 80.01% to 81.77%. The gains do not establish significance or double descent.

When correction children help and when they harm

Dataset / augmented reference	Root mistakes fixed	Correct predictions spoiled	Net images gained
CIFAR, 5,000 validation images	380	297	83
Cats, 2,326 validation images	92	99	-7

We compare the final predicted class against the frozen root on the same images. Corrections can fix real root mistakes while also firing on examples the root already classified correctly. On CIFAR the balance is positive; on Cats it is slightly negative. This explains why memorization alone is not enough, and gives a direct interpretation of the tree beyond a single accuracy number.

The views are correlated transformations of the same images. Only one deterministic validation-view bank was measured. Voting costs eight image evaluations per model instead of one; it adds no trained parameters. Its benefit is a measured prediction-time tradeoff, not a claim that more computation always helps.

7. Focused noisy-subset correction-width study

We trained depth-one correction trees on 4,000 balanced CIFAR images (400 per class) with an inherited 15% noise map (566 labels changed). The CNN/root stayed frozen. Widths 8-2048 were tested with original inputs or three fixed views. Each child had 1,000 epochs, with no weight decay, dropout or label smoothing. Interpolation means zero errors on every fitting view.

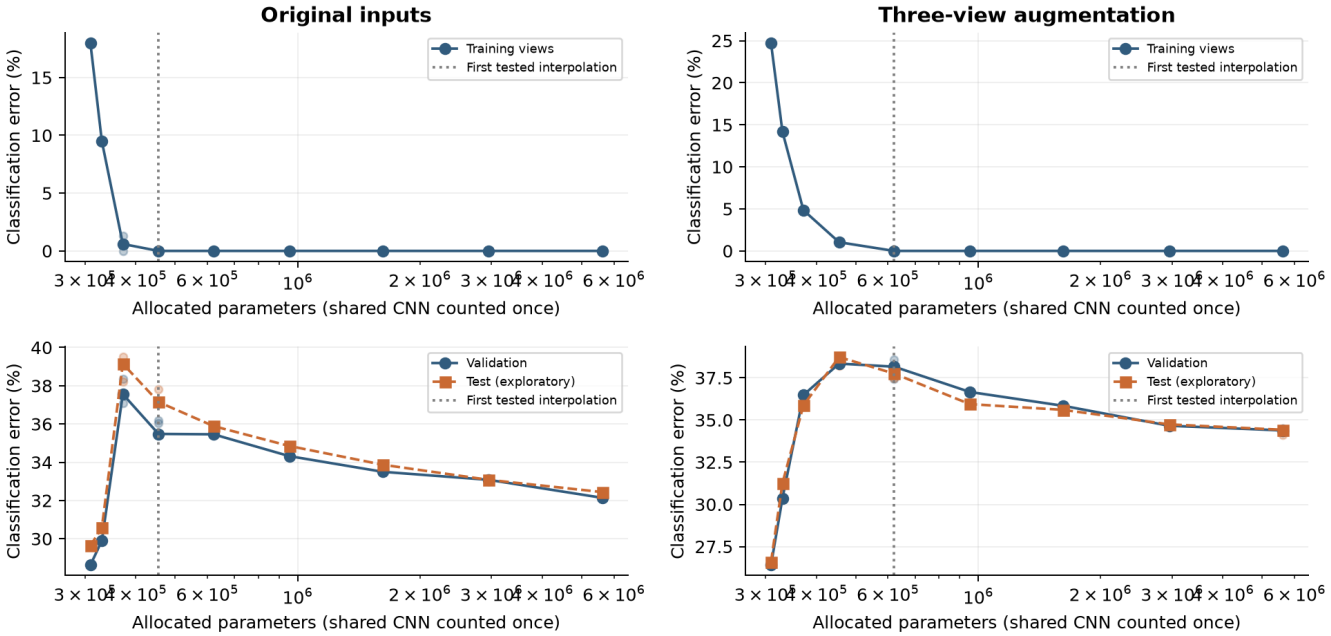


Figure 8. Noisy-subset correction capacity. Main lines use seed 409; faint points are additional optimizer seeds. Vertical lines mark the first tested zero-error model, where one exists. Missing widths are not interpolated experimental results.

Inputs	Primary widths completed	First zero-error width	Zero-error widths tested
Original	9	64	6
Augmented	9	128	5

Inputs	First zero-error width	Peak test error	Error at 2048	Gain
Original	64	38.95% (w=32)	32.32% (w=2048)	6.63 pp
Augmented	128	38.53% (w=64)	34.30% (w=2048)	4.23 pp

Both arms show a second descent in the sampled curve. With original inputs, the three-seed test error peaks at 38.95% (width 32), just before first interpolation at width 64, then falls to 32.32% at width 2048. With augmentation, it peaks at 38.53% (width 64), near first interpolation at width 128, then falls to 34.30%. All three seeds at the peak and widest widths agree on the decline.

This supports double descent from tree capacity in this noisy 4,000-image setup; the full-split study did not establish it. The subset uses one noise map and a previously used test split, and its frozen CNN saw the full clean training pool, so treat the finding as exploratory.

8. Conclusions and next steps

What worked. Fixed image-view banks made augmentation compatible with hard recursive correction tasks. The repetition control separated transformations from extra exposure. Frozen roots made added tree learning visible. Grouped bootstraps served both majority voting and bias-variance analysis, and gate-flip maps made the landscape limitations explicit.

What the evidence supports. The tree can memorize beyond an imperfect CNN root, but perfect fitting alone does not ensure better test accuracy. The full-data study did not establish double descent within its tested range. A separate noisy 4,000-image CIFAR subset shows a pronounced double-descent pattern in test error versus correction width: the peak occurs near interpolation and test error falls at wider capacities in both input settings. The direction repeats across three optimizer seeds at the sampled peak and widest widths. This focused result is exploratory because it reuses the earlier test split and one fixed noise map.

Limits of the full-data study. Widths above 512 were not tested on the full training splits; some largest noisy models still fail to interpolate, and several nodes have continuing loss reduction at the budget limit. The separate subset extension changes the number of training originals. Depth growth can stop before its requested limit. CNN/noise-map randomness was not independently repeated, and previously used test partitions make these findings exploratory. Shared pretrained features also limit the interpretation of sample-size experiments.

Next experiment. Keep the root, finite input bank and noise map fixed, locate the interpolation region using training errors, then test several genuinely larger correction widths with adequate convergence and independent seeds. For training-time double descent, collect dense complete-classifier checkpoints instead of relying only on three per-node budget snapshots. Use validation to plan the extension, then evaluate a fresh holdout.

Glossary

Term	Meaning in this report
Width	Hidden units in one correction MLP.
Epoch	One pass through a child's assigned training task.
Interpolation	Zero classification errors on all actual fitting targets/views.
Double descent	Held-out error improves, worsens near interpolation, then improves again.
Bootstrap / ensemble	Resample original images with replacement; combine final class votes.
Bias proxy / variance	Mean prediction's squared error against observed labels / spread across members.

Short bibliography

[1] Course materials: *Tree type network revisited*; *Tree type NN - exploring double descent and other aspects*; supplied error-correction-tree teaching manuscript.

[2] Nakkiran et al. (2019), *Deep Double Descent: Where Bigger Models and More Data Hurt*. arxiv.org/abs/1912.02292

[3] Breiman (1994), *Bagging Predictors*, Technical Report 421, UC Berkeley. Original report.

[4] Li et al. (2018), *Visualizing the Loss Landscape of Neural Nets*, NeurIPS. arxiv.org/abs/1712.09913

Code, measured results and reproduction instructions: github.com/Harshal-Narsaria/ELL409. Prior reports and full historical evidence are retained separately. The contribution is the controlled adaptation and experimental analysis, not a claim of a new general learning algorithm.